

Supplementary note: Bayesian comparison of HMG and MOND on galaxy rotation curves

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This note is supplementary material to Monjo & Banik (2025), *Distinct acceleration relations of galaxies and galaxy clusters from hyperconical modified gravity*, ApJ **992**, 35 [1].

Abstract

We test whether the neighbourhood parameter ε_H of hyperconical modified gravity (HMG) must be fitted per galaxy or is set by the local baryonic density. It is set by the density through a single global scale s (best-fit $s = 4.06$, recovering the $s = 4$ of the main text): one parameter, not one or two per galaxy. On the high-quality McGaugh (2007) sample this one-parameter model reaches $\chi^2_\nu = 1.47$ against 3.15 for MOND and is preferred by the Bayesian information criterion (BIC); the large information-criterion penalties quoted for HMG apply to a per-galaxy configuration the theory does not require. We do not claim outperformance of MOND: on the full SPARC sample fixed baryons are inadequate for either model, and with per-galaxy M/L fitting the two are at parity.

1 Method

We use the McGaugh (2007) high-quality galaxy sample as compiled for the main analysis [1] and the galaxy-cluster radial-acceleration sample (10 clusters, 108 points). The compilation has 61 galaxies with finite parameters (696 points), three of them with a single measured point; the 60 high-quality curves quoted in the main text omit one such galaxy (UGC 6923, dropped from its Fig. 5), and restricting to those 60 leaves every conclusion below unchanged. The baryonic circular speed $V_{\text{bar}} = (V_\star^2 + V_{\text{gas}}^2)^{1/2}$ is held fixed for every model, so all comparisons are made on identical baryons with no stellar mass-to-light fitting. This fixed-baryon comparison is meaningful only where the stellar M/L is well determined: on the full SPARC sample it is not (both models then give χ^2_ν of 20 to 34, a cross-check included in the reproduction script), so per-galaxy M/L fitting would be required there, under which HMG and MOND reach parity (as in the Wilcoxon test of Aydogdu 2026). We therefore restrict to the high-quality sample.

HMG predicts the circular speed through

$$V^2 = V_{\text{bar}}^2 \left[1 + \frac{2c/t}{a_N \gamma_0} \right]^{1/2}, \quad \gamma_0 = \frac{\gamma_{\text{sys}}}{\cos \gamma_{\text{sys}}}, \quad \sin^2 \gamma_{\text{sys}} = \sin^2 \gamma_U + (\sin^2 \gamma_{\text{cen}} - \sin^2 \gamma_U) q, \quad (1)$$

where $a_N = V_{\text{bar}}^2/r$ is the Newtonian baryonic acceleration, $q = |u^2 - \varepsilon_H^2|/(u^2 + \varepsilon_H^2)$ is the neighbourhood weight with $u = \sqrt{2} V_N/V_H$ and $V_H = r/t$, and γ_{sys} runs between the empty-space angle $\gamma_U = \pi/3$ and the central angle $\gamma_{\text{cen}} = 0.48\pi$ as the neighbourhood parameter ε_H varies. In the *general model* ε_H is not fitted: it is set by the local baryonic density [2],

$$\varepsilon_H^2(s) = \frac{\rho_{\text{nei}}(s)}{\rho_{\text{vac}}} + \frac{1}{6}, \quad \rho_{\text{nei}}(s) = \frac{M_{\text{bar}}}{\frac{4}{3}\pi(s r_{\text{sys}})^3}, \quad \rho_{\text{vac}} = \frac{3}{8\pi G t^2}, \quad (2)$$

with $s = 4$ for galaxies. Since $M_{\text{bar}} = V_N^2 r/G$, the density term is identically $\rho_{\text{nei}}/\rho_{\text{vac}} = 2(V_N/V_H)^2/s^3 = u^2/s^3$; ε_H is therefore set by the observed mass and size once the scale s is chosen. We calibrate s globally to the galaxies, a single fitted parameter; its best fit $s = 4.06$

recovers the $s = 4$ of the main text. This scale is the only quantity fitted: γ_{cen} is fixed and no parameter is fitted per galaxy, against MOND’s single fixed a_0 .

MOND [3] predicts $V^2 = V_{\text{bar}}^2 \nu(a_N/a_0)$ with the simple interpolating function and $a_0 = 1.2 \times 10^{-10} \text{ m/s}^2$ held fixed.

The galaxy data carry no per-point velocity errors. As a representative proxy we combine in quadrature the two median summaries of the SPARC $e_{V_{\text{obs}}}$ distribution (Lelli et al. 2016): the median fractional error, 4.3%, and the median absolute error, 4.6 km/s, giving $\sigma_V = [(0.043 V_{\text{obs}})^2 + (4.6 \text{ km s}^{-1})^2]^{1/2}$. This is a conservative sensitivity choice, not the published per-point error, and the conclusion does not rest on it: the error-model-free BIC_p below favours the one-parameter model equally ($\Delta\text{BIC}_p = -261$). We quote $\chi^2 = \sum [(V_{\text{model}} - V_{\text{obs}})/\sigma_V]^2$ with the Bayesian information criterion $\text{BIC} = \chi^2 + k \ln N$ [6], where N is the number of points and k the number of fitted parameters. Independently, we also report an error-model-free selection: the profiled-variance BIC on the $\ln V$ residuals, $\text{BIC}_p = N \ln(\text{SSR}/N) + k \ln N$. Its variance parameter is common to all models and cancels in ΔBIC_p ; the two likelihoods differ, so BIC and BIC_p are compared only within their own column. Clusters use the tabulated acceleration errors directly.

2 Results

Galaxies. Table 1 lists the fits. With ε_H set by the density through a single global scale s (best-fit 4.06), HMG reaches $\chi_\nu^2 = 1.47$ against 3.15 for MOND. Both exceed unity ($p = 1 \times 10^{-14}$ and 3×10^{-154}): with baryons held fixed a reduced χ^2 above one is expected, and only the many-parameter fits reach $p \gtrsim 0.4$. The comparison is therefore relative; by the BIC the one-parameter model is preferred over MOND ($\Delta\text{BIC} = -1167$; $|\Delta\text{BIC}| > 10$ counts as decisive on the scale of [7]). Adding per-galaxy freedom does not change the picture against MOND: the 122-parameter fit (the analogue of the penalised 248) is indistinguishable from the one-parameter model in the χ^2 -based BIC (1021 versus 1025), although the error-model-free BIC rewards its extra flexibility. The density-predicted and best-fit ε_H correlate at $r = 0.88$, consistent with ε_H being set by baryonic structure. This is a statement of parameter economy on a sample where fixed baryons are valid, not of universal outperformance of MOND.

Table 1: Galaxy rotation-curve fits (McGaugh 2007; 61 galaxies, 696 points; identical fixed baryons). ΔBIC is against MOND with the SPARC errors; ΔBIC_p is the error-model-free profiled-variance form. Negative values favour the row over MOND.

Model	params	k	χ^2	χ_ν^2	BIC	ΔBIC	ΔBIC_p
HMG, global density scale s (best-fit 4.06)	1	1	1018	1.47	1025	−1167	−261
HMG, ε_H per galaxy	61	61	641	1.01	1040	−1152	−225
HMG, ε_H , γ_{cen} per galaxy	122	122	222	0.39	1021	−1171	−535
MOND (a_0 fixed)	0	0	2192	3.15	2192	0	0

Clusters. The per-cluster scale s (Table 2) is a fitted diagnostic, one value per cluster, not an independent prediction: it shows only that the ten clusters lie on the density relation (2) for scales of order unity (median $s = 0.91$, range 0.37 to 2.11). Clusters are the regime where MOND itself requires additional unseen mass.

Data availability

The analysis of Tables 1 and 2 is reproducible with `supplement/bayesian_comparison.R` at https://github.com/robertmonjo/distinct_acceleration_relations.

Table 2: Best-fit $\varepsilon_H^{\text{fit}}$ (general model, $\gamma_{\text{cen}} = \pi/2$) and the fitted per-cluster scale s placing each cluster on the density relation (2). Here s is a diagnostic, not a prediction.

cluster	A0085	A1795	A2029	A2142	A3158	A0262	A2589	A3571	A0576	A0496
$\varepsilon_H^{\text{fit}}$	39.5	55.5	52.0	46.0	53.5	95.5	60.0	57.5	43.5	50.5
s	1.52	0.37	2.11	1.28	1.01	0.42	0.73	0.82	0.60	1.24

References

- [1] Monjo R., Banik I., 2025, ApJ, **992**, 35.
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- [7] Kass R.E., Raftery A.E., 1995, J. Am. Stat. Assoc., **90**, 773.